Tech News

Wearable devices powered with energy harvested from radio waves

A Penn State University-led research team has developed a way to harvest energy from radio waves to power wearable devices. With the help of a stretchable wideband dipole antenna system capable of wirelessly transmitting data, the system retains its frequency functions (even when stretched, bent, or twisted) and converts energy from electromagnetic waves into electricity using a rectified antenna, or 'rectenna.' This electricity can then be used to power wireless devices or charge energy storage devices such as batteries and supercapacitors.

An international team of researchers, led by Huanyu "Larry" Cheng, Dorothy Quiggle Career Development Professor in the Penn State Department of Engineering Science and Mechanics, has developed a stretchable antenna and rectenna system that harvests energy from radio waves in the ambient environment to power wearable devices (Credit: https://news.psu.edu)

Impressive ultra-thin sensor suitable for health monitoring

Long-term exposure to nitrogen dioxide (NO_2) is known to cause severe heart and respiratory diseases such as asthma and infections. As it is odourless and invisible, a special sensor is required to detect its hazardous levels. However, most currently available sensors are energy-intensive for achieving suitable performance. Now a team of researchers from the Berkeley Lab and UC Berkeley have devised an ultra-thin sensor that can work at room temperature and thus consume less power than conventional sensors. The new 2D sensor electrically responds selectively to NO_2 molecules, with minimal response to other toxic gases such as ammonia and formaldehyde. Once detected, the sensor recovers to its original state within a few minutes.

The new 2D sensor is flexible and transparent, making the technology a likely candidate for wearable environmental-and-health-monitoring sensors (Credit: https://newscenter.lbl.gov)

Nanotransistors that stay cool even at high voltages

Power converters are devices responsible for powering various electrical/electronic equipment. Heat dissipation in converters caused by the high electrical resistance, among other factors, is a big challenge in power electronic devices, decreasing the range of electric vehicles and efficiency of renewable-energy systems. Now a research team at EPFL has developed a transistor that can substantially reduce the resistance and cut the amount of heat dissipation in high-power systems. More specifically, it has less than half as much resistance as conventional transistors, while holding voltages of over 1,000V.

Nanotransistors that stay cool at high voltages (Credit: https://actu.epfl.ch)